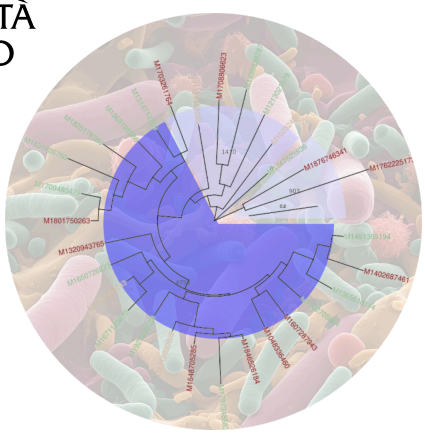
Microbial Genome Analysis in Healthy and Disease-Affected Human Oral Environments



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Introduction

Human oral cavity hosts a diverse microbial ecosystem, with over 700 identified bacterial species contributing to both oral and systemic health. These communities play key roles in immune regulation, nutrient metabolism, and pathogen defense (Dewhirst et al., 2010). However, disruptions to this balance, known as dysbiosis, can lead to various pathological conditions, including peri-implant mucositis and peri-implantitis (Heitz-Mayfield, 2024). Understanding the genomic and functional changes associated with these imbalances is critical for improving treatment strategies. Recent advances in high-throughput sequencing and metagenomic pipelines have enabled recovery of such data directly from complex microbial communities with metagenome-assembled genomes (MAGs). This approach allows for detailed exploration of taxonomic profiles and metabolic potential of microbes without time-consuming cultivation (Almeida et al., 2019).

The aim of this study is to characterize microbial genomic content from individuals with healthy oral conditions and those affected by mucositis and peri-implantitis. Analysis of the genome completeness, functional genes, metabolic pathways, and phylogenetic relationships is performed to elucidate the microbial dynamics associated with oral health and disease.

Methods

Data overview

Data used in this study consisted of 30 metagenome-assembled genomes (MAGs), recovered after sequencing dental plaque samples of 30 individuals (3 “healthy”, 10 “mucositis”, and 17 “peri-implantitis”). In addition to the study group, MAGs were supported with metadata, describing the age, sex, body mass index, and smoking state (non-smoker, smoker, ex-smoker) of each individual.

Quality checking

CheckM (*v.1.2.3.*) was used to assess the quality of the binned MAGs (*taxonomy_wf*). Both technical and reference-based methods with specific marker genes were applied to evaluate the integrity of genomes, with key metrics including genome completeness, contamination, N50, GC-content, genome size, and others. MAGs were classified into quality categories based on established thresholds: low-quality samples (completeness < 50% or contamination > 5%), medium-quality samples (50% ≤ completeness < 90% and contamination ≤ 5%), and high-quality samples (completeness ≥ 90% and contamination ≤ 5%).

Taxonomic analysis

Taxonomic classification of MAGs was carried out using *PhyloPhlAn Metagenomic* (*v.3.0.36*).

MAGs were assigned to species-level genome bins (SGBs) based on the computed average *Mash* distance between each MAG and reference genomes. MAGs with average Mash distance < 0.05 to a reference genome were assigned to the corresponding SGB.

Genome annotation

Prokka (*v.1.14.6*) was employed to annotate genomic features of the contigs within bins. For each MAG, multiple output files were generated, detailing genomic features such as coding sequences (CDS), non-coding transcripts, repetitive elements, transfer RNAs (tRNA), ribosomal RNAs (rRNA), and predicted proteins. General Feature Format (GFF) files were visualized with Simple GFF Viewer (http://gff.molgenrug.nl/).

Fisher's exact test was applied to each annotated gene to identify significantly differing presence between healthy and disease groups (Upton, 1992). Genes with p-value < 0.05 were classified as differentially present.

Metabolic annotation

Metabolic potential of MAGs was characterized using *KEGG Automatic Annotation Server* (KAAS): (https://www.genome.jp/tools/kaas/).

Predicted protein sequences (FASTA) obtained from *Prokka* were merged inside each study group, and used as input. The KAAS annotation was performed using the *bi-directional best hit* (*BBH*) method, with GHOSTX as the search algorithm and a threshold of 60.

Pan-genome analysis

Roary (*v.3.13.0*) was utilized in pan-genome characterization to assess the shared gene content and genetic diversity across MAGs. Protein sequences from GFF files were used for more stable alignment, with *BLASTp* minimum identity threshold of 95%. The pan-genome analysis sorted genes in 4 groups based on their presence: core genes in > 90% of bins; soft-core genes in 89% of bins; shell genes in ≥ 15% and < 89% of bins; cloud genes in < 15% of bins.

Post-analysis, R and Python scripts (https://github.com/sanger-pathogens/Roary) were used to generate visual plots, including pan-genome pie charts, presence/absence graphs, gene clustering diagrams, and others.

Phylogenetic analysis

Accessory gene phylogeny was inferred from the *Roary* binary presence/absence matrix, while core gene phylogeny was reconstructed based on nucleotide alignments of core genes using *FastTreeMP* (*v.2.1.11*). This approach employed a *generalized time-reversible* (*GTR*) model for nucleotide substitution and maximum likelihood estimation, incorporating mutation rate variation across sites.

Data processing and visualization

Output data manipulation and visualization was performed using Python (*pandas*, *numpy*, *matplotlib*, *seaborn*) and R (*dplyr*, *tidyverse*, *ggplot2*) libraries. Phylogenetic trees were analyzed with R (*ggtree*, *ape*, *phytools*) packages, using metadata annotations for group differentiation. Midpoint rooting was applied to the resulting phylogenies to estimate the root position.

All supplementary materials are accessible with the following link:
https://tinyurl.com/alshaarbertova

Results and Discussion

To characterize microbial genomic content, quality checking was performed using *CheckM*, which assessed completeness and contamination metrics by comparing MAGs with a reference database of single-copy marker genes, alongside contig statistics analysis. Based on quality parameters, MAGs were classified into two groups: 15 high-quality and 15 medium-quality, with no

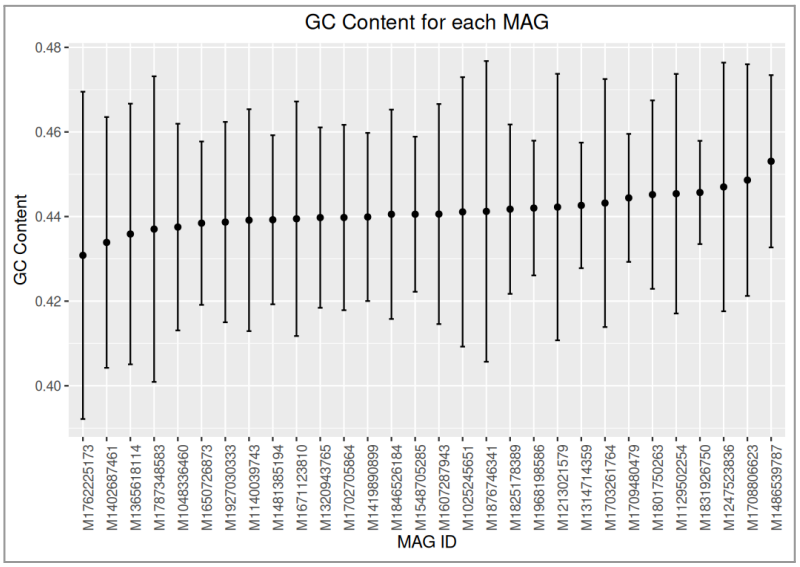
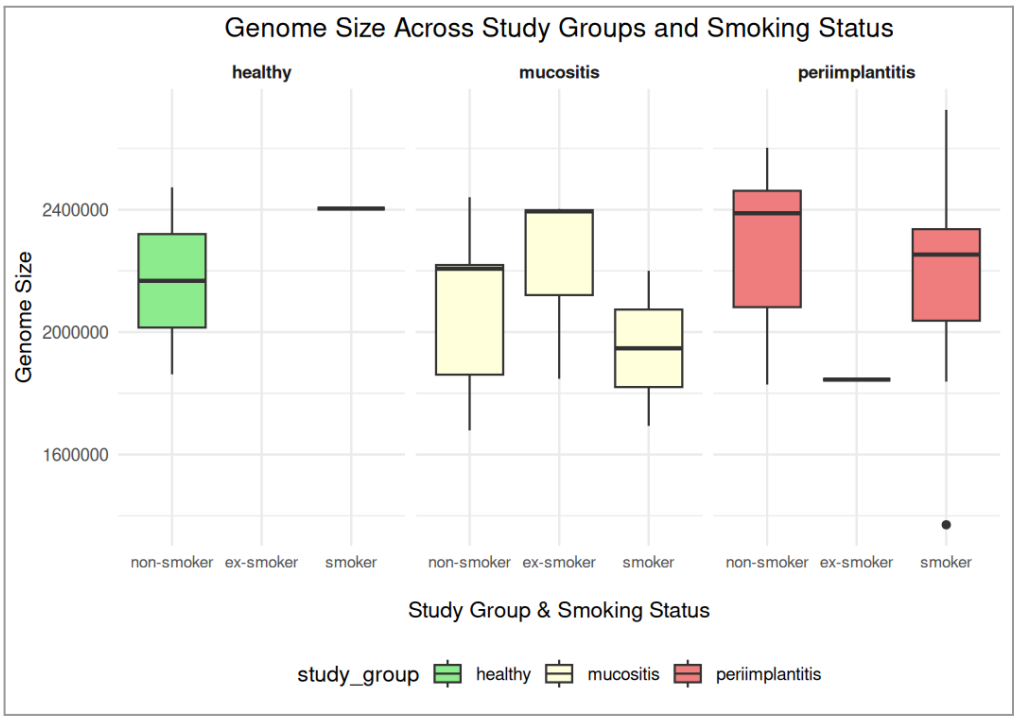


Fig. 1. Average GC content among the contigs of each MAG (error bars depict standard deviation)

Fig. 2. Genome sizes (bp) among MAGs of different study groups (healthy, mucositis, peri-implantitis)



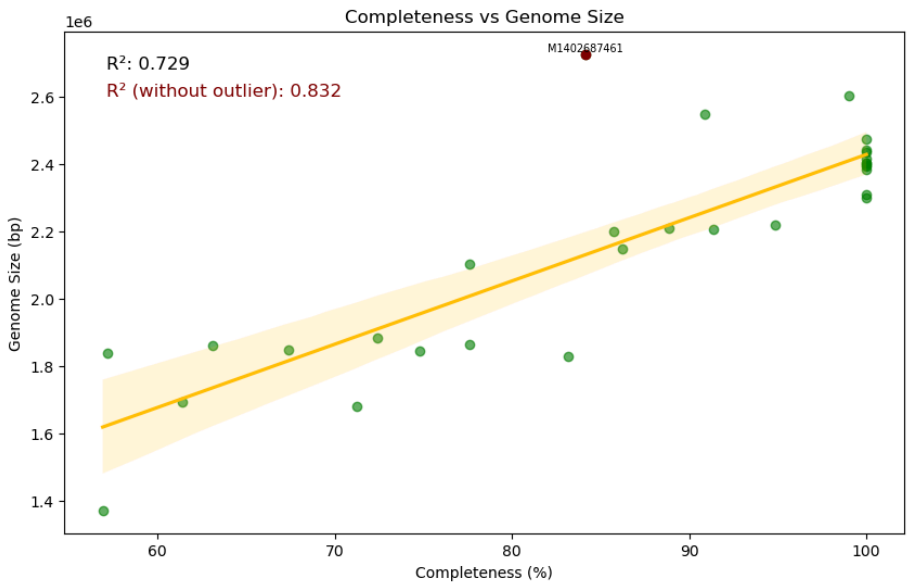


Fig. 3. Linear regression analysis of the genome size (dependent var.) vs. genome completeness (independent var.), with $R^2 = 0.729$, excluding outlier M1402687461 with $R^2 = 0.839$

low-quality samples identified (**Suppl.**). GC content analysis in genomic bins revealed an average range of 43% - 45% (**Fig.1.**). Plotting the GC content and its standard deviation (SD) across all contigs in each MAG indicated no significant differences in GC content among the 30 genomic bins, with sequencing quality as a potential source of variation. In contrast, genome size exhibited a broader range of 1.37 - 2.72 Mbp, with peri-implantitis samples showing the widest variation (1.37 - 2.7 Mbp), and healthy patient samples having a more consistent genome size range (1.86 - 2.47 Mbp) (**Fig.2.**). This discrepancy may be attributed to both the larger number of peri-implantitis MAGs (17 samples) compared to the healthy group (3 samples) and the possibly greater diversity in the disease-associated microbial communities.
Linear regression analysis was performed to investigate the relationship between completeness across the MAGs and their size (**Fig.3.**). A correlation between completeness and genome size was observed, with bin *M1402687461* identified as an outlier (> 1.96 SD from the regression line). This MAG, derived from a female peri-implantitis patient (smoker), exhibited 84.12% completeness and genome size of 2.72 Mbp, 23% larger than the median. Despite relatively low 3.45% contamination, it contained 2,626 predicted genes, notably higher than the dataset's median (1,972.5). Assembly quality metrics indicated an N50 of 14,563, considerably lower than the dataset's median (76,426.5). Possible explanations include strain-level genomic variation, assembly fragmentation due to the low N50, or redundant contigs inflating the genome size.

To determine the microbial composition, taxonomic analysis was performed using *PhyloPhlAn Metagenomic*, specifying Bacteria as the input domain. The results revealed a high degree of taxonomic similarity among the recovered MAGs, with all genomes exhibiting an average *Mash* distance < 0.05 when compared to SGB1580. Low dissimilarity suggested a close evolutionary relationship, leading to assigning all the bins to *Prevotella koreensis* species (**Suppl.**).
P. koreensis is a Gram-negative, rod-shaped, anaerobic bacterium first isolated in 2019 from a dental plaque of a female periodontitis patient in Korea (Park et al., 2019). *Prevotella* spp., commonly found in the oral, respiratory, gut, skin, and vaginal microbiomes, are implicated in various conditions like periodontitis, peri-implantitis, abscesses, vaginal and bone infections, and many others (Tett et al., 2021). In peri-implantitis, *Prevotella* spp. dominate peri-implant pockets, playing a critical role in biofilm formation, tissue inflammation through immune responses, and synergistic interactions with other pathogens, ultimately disrupting oral health.

To annotate genomic features in *P. koreensis* strains, the total number of genetic elements and coding sequences (CDS) in MAGs was assessed using *Prokka*. Results revealed a range of 1,278 - 2,463 genes per genome, with considerable variation in gene content and the proportion of hypothetical proteins across samples (**Fig.4.**). The number of CDS generally mirrored the total gene count, reflecting a predominance of protein-coding genes. However, a substantial fraction (40–60%) of predicted proteins were annotated as hypothetical, indicating uncharacterized functions. Notably, strain *M1402687461* showed a higher proportion of hypothetical proteins (1500) than others. Larger genome and higher number of hypothetical proteins in *M1402687461* may indicate unique genomic features worth further investigation.
Fisher's test revealed 17 genes that were significantly different between the healthy and disease groups (mucositis + peri-implantitis), with all but two of these genes classified as unknown (**Fig.5.**). The identified genes were *pqqE*, involved in redox homeostasis, and *mnmE*, which controls the modification of tRNA molecules.

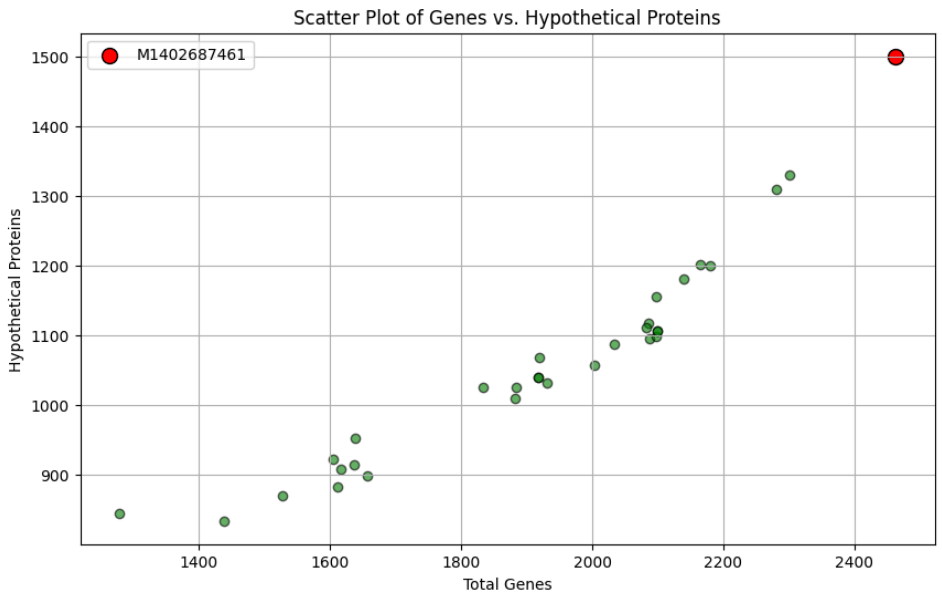
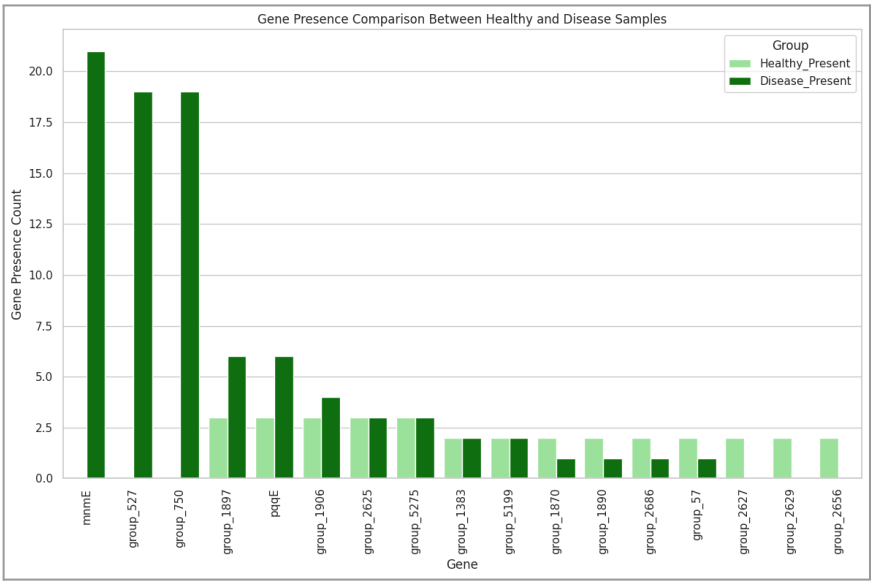


Fig. 4. Scatter plot of the number of hypothetical proteins vs the number of genes in *Prevotella koreensis* strains. Strain *M1402687461* was characterized by the largest amount of predicted protein-coding genes

Fig. 5. Gene presence comparison between health and disease-associated (mucositis + peri-implantitis) *Prevotella koreensis* strains



Gene	Protein	Function_in_Bacteria	KEGG_Bacteria	Human_Homolog	KEGG_Humans
mnmE	MnmE, tRNA modification GTPase	tRNA modification enzyme	GidA/MnmE virulence	GTPBP3	Transfer RNA biogenesis
ppqE	PqqE, coenzyme PQQ biosynthesis protein E	biosynthesis of a redox cofactors	PQQ biosynthesis pathway	-	-
mdr1	Mdr1, multidrug resistance protein 1	efflux pump for toxins, drugs, xenobiotics	ABC transporters, Drug resistance	ABCB1 (P-glycoprotein)	Gastric cancer
msh2	Msh2, DNA mismatch repair protein	DNA mismatch repair system	Mismatch repair	MSH2	Colorectal cancer
auaC	AuaC, polyketide synthase	biosynthesis of polyketide antibiotics	Biosynthesis of antibiotics	-	-
hprT	HprT, hypoxanthine phosphoribosyltransferase	hypoxantine-mediated purine salvage	Purine metabolism	HPRT1	Drug metabolism
rnc	Rnc, ribonuclease III	RNA maturation and decay	RNA degradation	DROSHA	Proteoglycans in cancer

Fig. 6. Proteins from disease-associated *Prevotella koreensis* strains (annotated by *Prokka* and *KAAS*), their human homologs and associated metabolic pathways

Previous studies have shown that the product of *mnmE* plays a crucial role in the virulence of *Streptococcus mutans*, a major contributor to dental caries (Li et al., 2013). MnmE, a multi-domain tRNA modification GTPase, has been implicated in bacterial survival under stress conditions, including acidic environments, high temperatures, and osmotic pressure, which are common in the oral cavity. Furthermore, the deletion of *mnmE* has been linked to a reduction in bacterial growth and an increased susceptibility to antibiotics, highlighting its essential role in pathogenicity. Although *mnmE* has been extensively studied in Gram-positive bacteria, its function in Gram-negative species such as *P. koreensis* remains largely unexplored. This raises the question of whether *mnmE* serves a comparable adaptive function in Gram-negative bacteria or whether other, unidentified factors contribute to the pathogenicity of *P. koreensis*.
The abundance of differentially present unknown genes between healthy and disease groups indicates complexity of microbial pathogenesis in mucositis and peri-implantitis. This suggests potential novel roles in bacterial survival and its association with oral diseases, emphasizing the need for further functional characterization.

Through the KAAS functional annotation of *Prokka*-derived protein sequences, multiple pathways were identified exclusively in disease-associated samples, including drug metabolism, biosynthesis of antibiotics (aurachin), and RNA degradation (**Fig.6.**). Notably, the bacterial Multidrug Resistance Protein 1 (MDR1; *EC*:7.6.2.2), a transporter implicated in xenobiotic efflux and antimicrobial resistance (van Veen et al., 1996), was detected in disease-associated *P. koreensis* strains. The presence of MDR1 suggests potential mechanisms for multidrug resistance and metabolic adaptation in oral microbial communities under inflammatory or pharmacological stress. However, it should be noted that the number of healthy samples was substantially lower than that of disease samples (3 against 27), which may have influenced the observed enrichment of disease-specific features.

To characterize strain diversity of *P. koreensis*, *Roary* pan-genome analysis was conducted. The approach categorized gene content into core, soft-core, shell, and cloud genes based on their presence across the studied genomes. With a default coreness definition of 90%, a total of 903 genes were classified as core genes (**Fig.7.A**). The shell gene category, with genes present in a substantial proportion of strains, comprised 1,410 genes, while the cloud gene group was the most abundant, with 4,758 genes present only in a minority of strains. These results indicate a diverse gene pool, with most of the gene content being accessory, reflecting high variability among strains. This gene distribution pattern suggests an open pan-genome of *P. koreensis*, typical of microbial species with a high degree of genetic plasticity (Tettelin et al. 2005). The relationship between the number of genomes analyzed and the number of conserved genes showed a clear trend: as the number of studied genomes increased from 1 to 30, the number of core genes gradually decreased (**Fig.7.E**). Conversely, the total number of genes in the pan-genome increased with the number of genomes studied, without reaching a plateau.
The high degree of genome variability suggests flexibility and adaptability of *P. koreensis*, likely influenced by its niche within the oral microbiome. Here, frequent genetic exchange between strains may occur due to environmental pressures and interactions with other microorganisms, contributing to genomic diversity.
The pan-genome matrix, based on the presence/absence of the genes, included 7,071 gene clusters (**Fig.7.B**). This matrix was then used to construct an accessory gene tree, which helped infer the phylogenetic relationships among the *P. koreensis* strains,

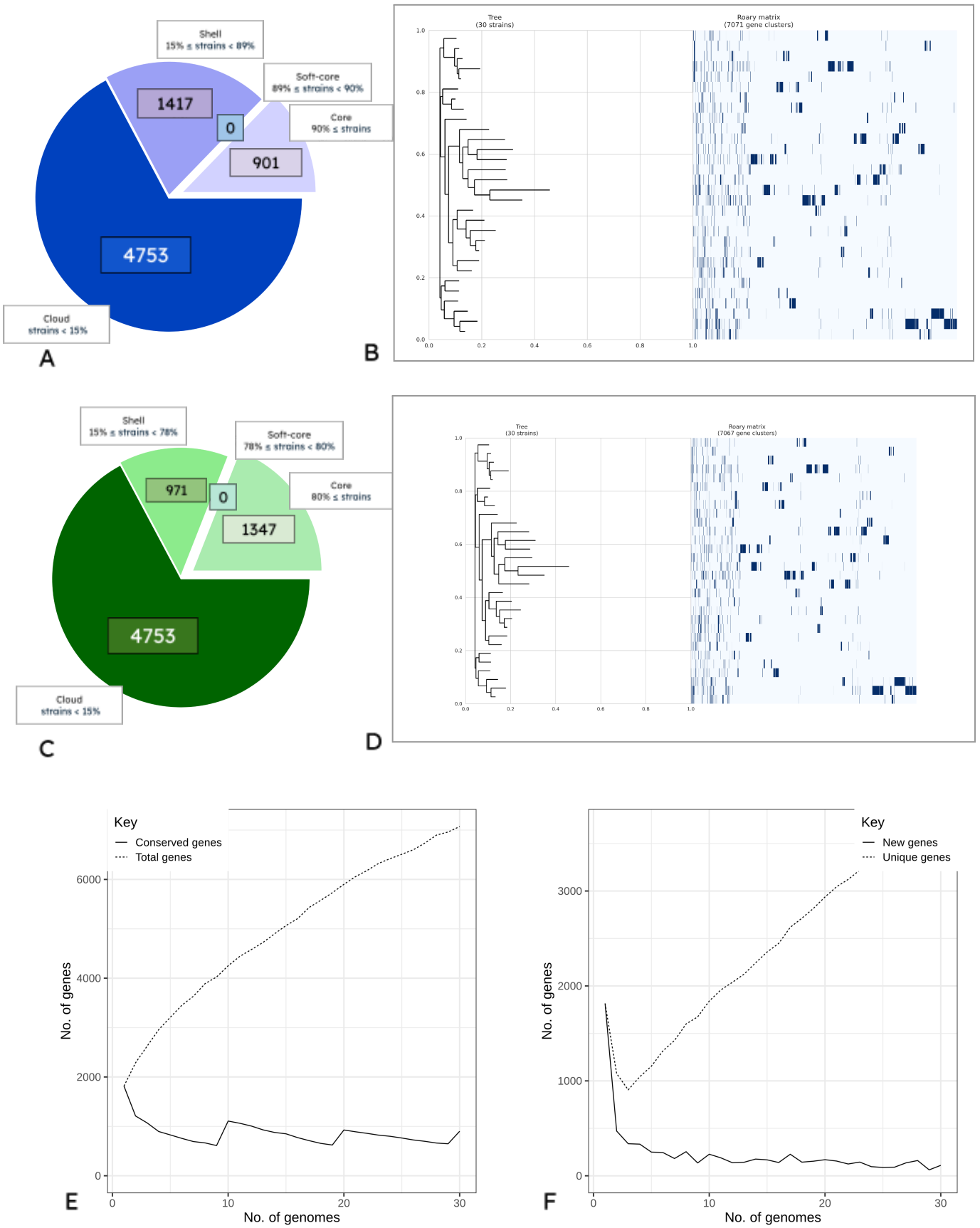


Fig. 7. Pan-genome composition pie-charts of the *Prevotella koreensis* strains with coreness definition of 90% (**A**), and 80% (**B**). Pan-genome matrix and identified gene clusters with core definition of 90% (**C**), and 80% (**D**). Line plots of the number of conserved genes vs. the number of total genes (**E**), and the number of new genes vs. the number of unique genes (**F**) in MAGS with coreness definition 90%

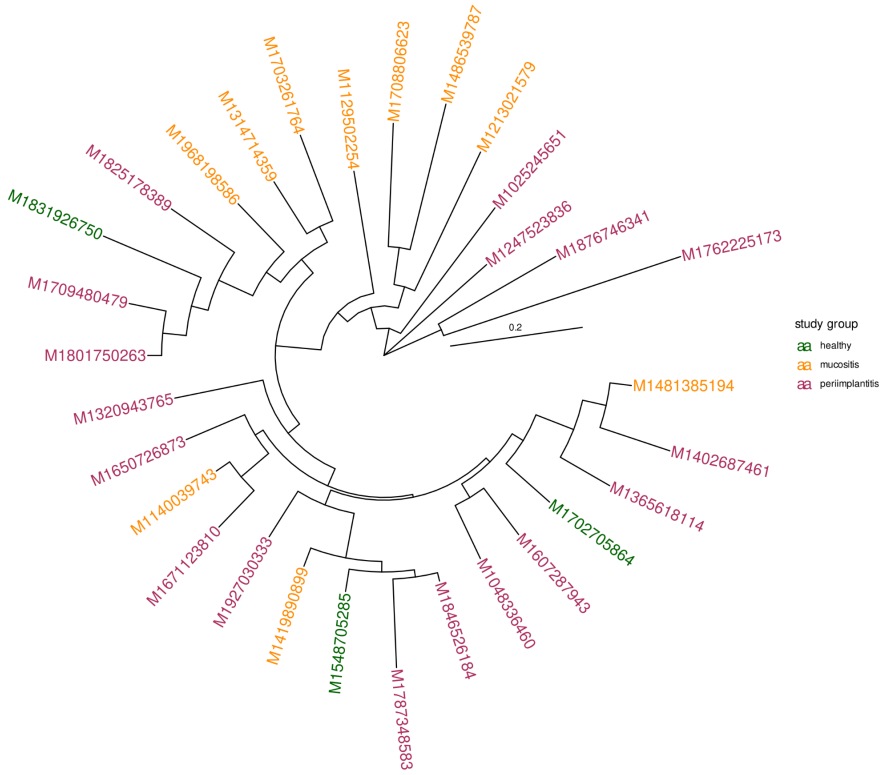


Fig. 8. *Roary* Phylogenetic tree of *Prevotella koreensis* strains, based on accessory gene presence/absence matrix (drawn with R *ggtree, phylotools, ape*). Tree is rooted at midpoint

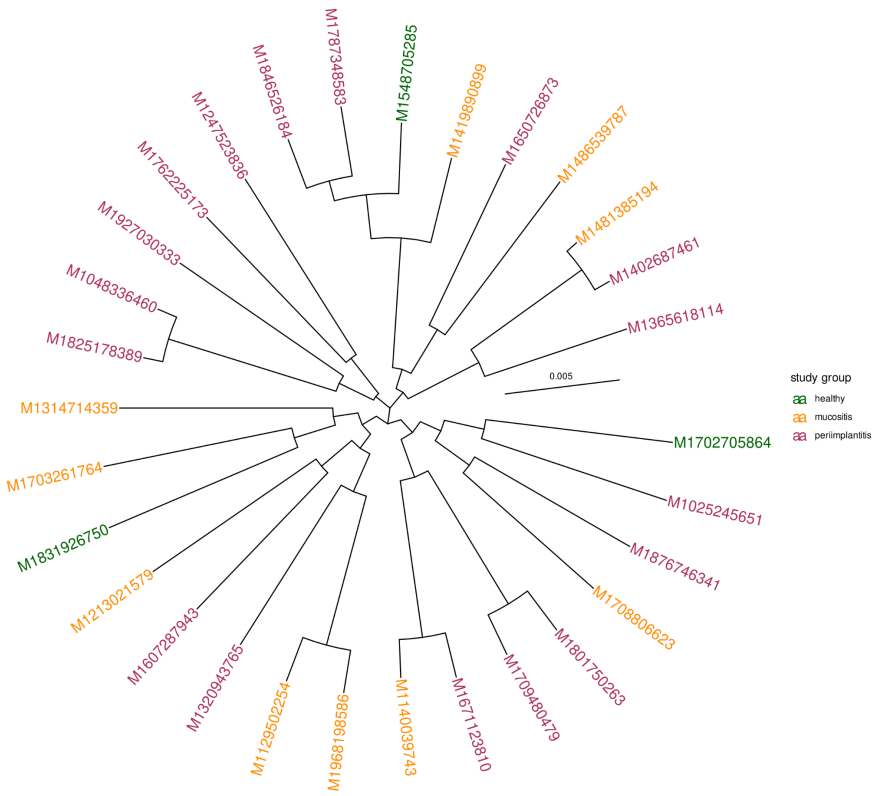


Fig. 9. *FastTree* phylogenetic tree of *Prevotella koreensis* strains, based on nucleotide sequence similarity of the core genes (drawn with R *ggtree, phylotools, ape*). Tree is rooted at midpoint

highlighting the diversity of accessory genes (**Fig. 8.**).

Phylogenetic analysis was performed in *FastTree* based on nucleotide sequences of the core genes derived from *Roary* (**Fig.9.**). The resulting core gene tree revealed no clear clustering across healthy, mucositis, and peri-implantitis samples, suggesting that the core genome is not directly influenced by disease state. In contrast, the accessory gene tree exhibited partial clustering, with some strains from peri-implantitis and mucositis environments grouping together, indicating that accessory genes are more influenced by the inflammation inside oral cavities. This suggests that while long-term evolution is not significantly shaped by disease states, the short-term is more influenced by environmental pressures, driven by mutations, gene acquisition or loss.

Annotation of both the core and accessory trees was also performed based on other host factors, such as sex, BMI, and age. Partial clustering was observed only in the accessory gene tree with annotation based on host sex, with no evident grouping observed for other factors (**Suppl.**).

Conclusion

This study characterized the genomic diversity and functional potential of *P. koreensis* in the variative human oral environments. The MAG M1402687461 suggested a possible strain-level divergence; however, higher-quality assemblies are needed to elucidate its potentially unique protein-coding profile. The open pan-genome and high accessory gene variability have been demonstrated, indicating the species' adaptability to the dynamic oral environment. This could include intra- and interspecies interactions, and adaptation to host factors such as inflammation or immune responses. The identified genes involved in disease-associated pathways, including antibiotic and multidrug resistance (*mnmE*, *mdr1*), highlight *P. koreensis*'s potential role in oral health and disease. Future research could examine the functional roles of accessory genes influenced by host context, particularly in diseases like peri-implantits and mucositis, and expand datasets to deepen insights into *P. koreensis* diversity.

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